UNIT-3

Error: -

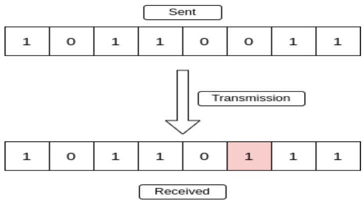
- if the data sent by the sender does not reach the receiver as intended and instead, some other data reaches the receiver, we call that a "error.

- A condition when the receiver's information does not match with the sender's information.
- During transmission, digital signals suffer from noise that can introduce errors in the binary bits travelling from sender to receiver.
- That means a 0 bit may change to 1 or a 1 bit may change to 0.

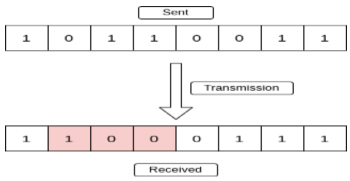
EG: - 100 change in 101 here one bit has been changed

There are two types of Error: -

1- **Single-Bit Error:** Single bit error means that only one bit in the data changes. In this case, a 0 can change to 1 or a 1 can change to 0.



2- **Burst Error:** Burst error means that two or more bits in the data change. In this case, 0s can change to 1s or 1s can change to 0s. In burst errors, the duration of noise is longer compared to single-bit errors.



Error detection

- **Error detection** is a technique used to identify hidden errors in data.

- In other words, "error detection" is a method employed to detect the presence of noise (interference) and data loss in data.

- Error detection minimizes the probability of passing incorrect frames to the destination, known as undetected error probability.

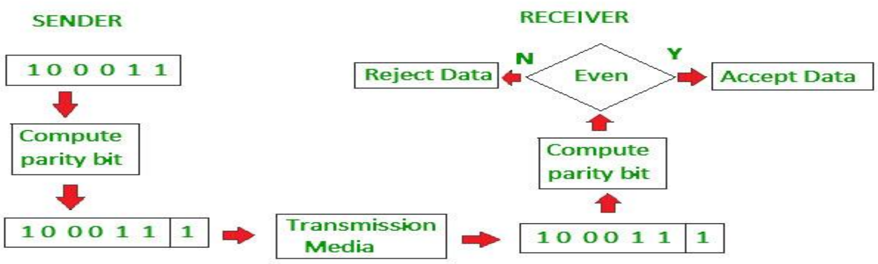
Error Detection Techniques are:

- **1- Simple Parity Check:** This is a method for detecting errors in data. It is effective in accurately identifying errors during communication.

- In the simple parity check method, a parity bit is added to the original data. Then, the data is transferred through the data link. If there is any corruption in the data, it can be detected.

- Blocks of data from the source are subjected to a check bit or parity bit generator form, where a parity of :

- 1 is added to the block if it contains odd number of 1's, and
- 0 is added to the block if it contains even number of 1's



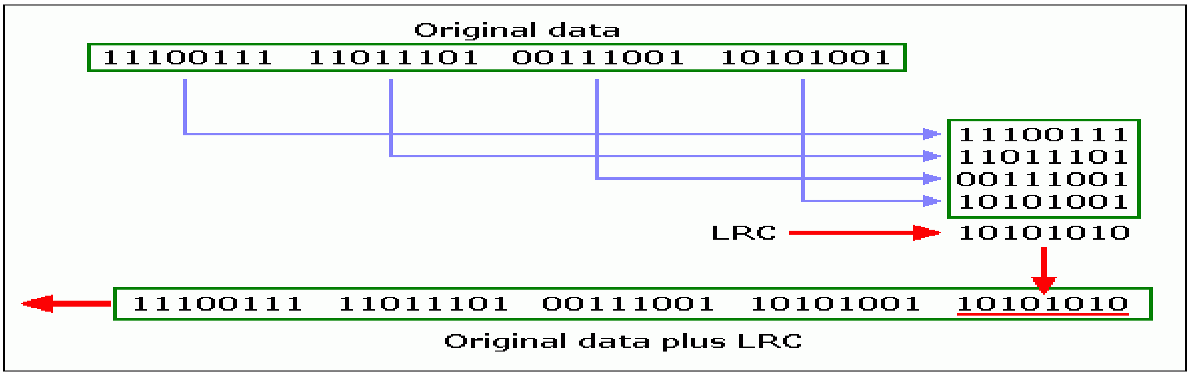
Limitations of Parity Checking:

- It is not suitable for detection of errors if the number of bits changed is even.
- If error is noticed, it cannot be corrected.
- It cannot reveal the location of erroneous bit.

2- Two-Dimensional Parity Check/ Longitudinal Redundancy Check (LRC):

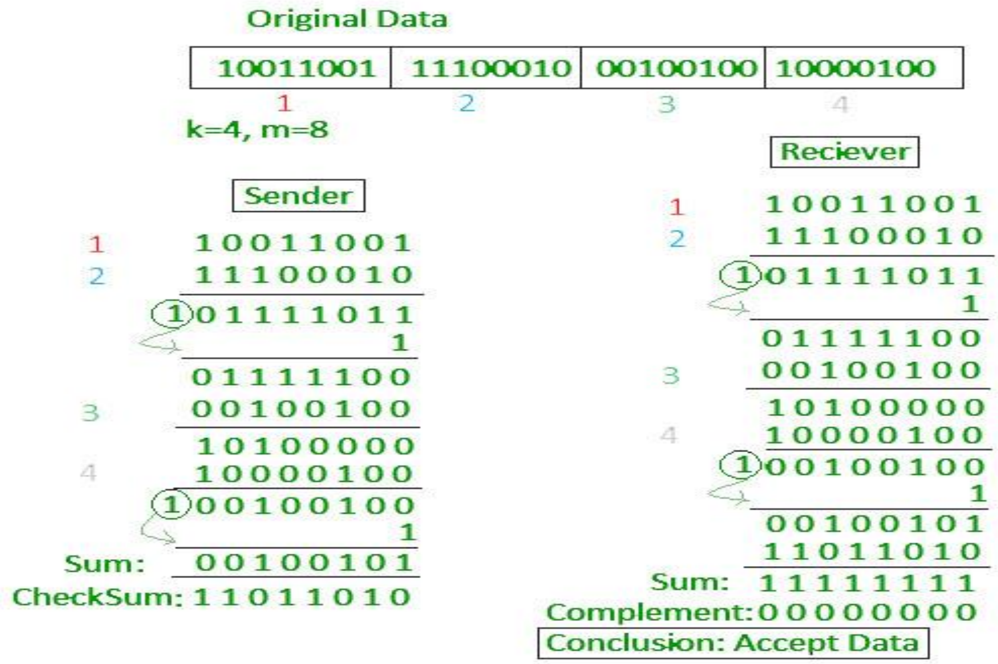
- This is a technique for detecting errors in which each line of data is checked. It is capable of detecting errors hidden in a larger amount of data and provides accurate data verification.

- LRC is an error-detection method for determining the correctness of transmitted and stored data.
- In this method, the block of bits is organized in a matrix or table of rows and columns.
- Then the parity bit for each column is calculated and a new row of eight bits, which are the parity bits for the whole block, is created.
- After that the new calculated parity bits are attached to the original data and sends to the receiver.



3- Checksum: Checksum technique is used to check for errors hidden in data. It ensures that there are no errors in the data. In this technique, a checksum segment is sent along with the data.

- In checksum error detection scheme, the data is divided into k segments each of m bits.
- In the sender's end the segments are added using 1's complement arithmetic to get the sum. The sum is complemented to get the checksum.
- The checksum segment is sent along with the data segments.
- At the receiver's end, all received segments are added using 1's complement arithmetic to get the sum. The sum is complemented.
- If the result is zero, the received data is accepted; otherwise discarded.



- **4- Cyclic Redundancy Check (CRC):** CRC, short for cyclic redundancy check, is a technique based on binary division. In this technique, bits are appended to the end of data units. CRC is commonly used for error-checking in data transmission and storage systems to verify the integrity of data.
- Unlike checksum scheme, which is based on addition, CRC is based on binary division.

- In CRC, a sequence of redundant bits, called cyclic redundancy check bits, are appended to the end of data unit so that the resulting data unit becomes exactly divisible by a second, predetermined binary number.
- At the destination, the incoming data unit is divided by the same number. If at this step there is no remainder, the data unit is assumed to be correct and is therefore accepted.
- A remainder indicates that the data unit has been damaged in transit and therefore must be rejected.

Error Correction Code

- Error Correction Codes are used to detect and correct errors in data. Whenever data is sent from a sender to a receiver, these codes are used to identify errors in the data and then correct them.
- In simple terms, "Error correction is a process through which errors in the transmitted data are identified and fixed."
- Error correction can be handled in two ways:
- **Backward Error Correction:** In this approach, once an error is detected, the receiver requests the sender to resend the entire data. This method ensures that the data received by the receiver is error-free by retransmitting the data.
- **Forward Error Correction:** In this method, the receiver uses error-correcting codes to correct errors in the received data. These error-correcting codes are designed in such a way that they can automatically correct errors in the data without the need for retransmission. This approach is more efficient in terms of bandwidth and latency because it doesn't require the sender to resend the data

These are some error correcting code below: -

- **Hamming Code:** This is a block code used to detect two-bit errors and correct single-bit errors. It achieves error detection and correction by adding redundant bits to the original data, allowing the receiver to identify and correct errors when they occur.
- **Binary Convolution Code:** This type of code involves an encoder that processes input sequences of arbitrary length and generates an output sequence (sequence of bits) based on a convolution operation. These codes are commonly used in digital communication systems.
- **Reed-Solomon Code:** Developed by S. Reed and Gustave Solomon in 1960, this is a block code designed to correct burst errors in received data blocks. It is widely used in applications where data integrity is crucial, such as data storage and transmission.
- **Low-Density Parity Check Code:** This is another block code specified by a parity-check matrix. It is particularly useful for sending data through very noisy

communication channels. LDPC codes are known for their excellent error correction performance and are used in various modern communication systems.

Framing:

- The breaking of bit stream by inserting spaces or time gaps is called framing.
- At data link layer, it extracts message from sender and provide it to receiver by providing sender's and receiver's address.
- Framing is a point-to-point connection between two computers or devices consists of a wire in which data is transmitted as a stream of bits.
- Framing is a function of the data link layer.
- It provides a way for a sender to transmit a set of bits that are meaningful to the receiver.
- Frames have headers that contain information such as error-checking codes.

Various Framing Techniques used for Framing are:

a. **Character Count:**
    - This method uses a field in the header to specify the number of characters in the frame.
    - When the data link layer at the destination sees the character count, it knows how many characters follow, and hence where the end of the frame is.

    Disadvantage of Character Count:

    - If the header field of the frame gets corrupted during transmission then the error can change the character count.
    - This drawback is difficult to overcome. Therefore, this method is not used now a days.

b. **Starting and ending character with Character Stuffing:**
    - In the second method, each frame starts with the ASCII character sequence DLE STX and ends with the sequence DLE ETX.(where DLE is Data Link Escape, STX is Start of TeXt and ETX is End of TeXt.)
    - This method overcomes the drawbacks of the character count method.
    - If the destination ever loses synchronization, it only has to look for DLE STX and DLE ETX characters.

c. **Starting and ending with Bit Stuffing:**
    - The third method allows data frames to contain an arbitrary number of bits and allows character codes with an arbitrary number of bits per character.
    - At the start and end of each frame is a flag byte consisting of the special bit pattern 01111110.

- Whenever the sender's data link layer encounters five consecutive 1s in the data, it automatically stuffs a zero bit into the outgoing bit stream.
- This technique is called bit stuffing.
- When the receiver sees five consecutive 1s in the incoming data stream, followed by a zero bit, it automatically de-stuffs the 0 bit.
- The boundary between two frames can be determined by locating the flag pattern

Flow control

Flow control is a technique used to control the flow of data. This technique also manages the flow of data.

- Flow control is an important function of the data link layer, through which the receiver is sent the correct amount of data.
- The primary goal of flow control is to prevent the overload of data sent by the sender, ensuring that data is not lost or damaged in any way.

1- **Stop and Wait Flow Control** This is the simplest method of flow control. In this method, data and information are divided into frames, and one frame at a time is sent to the receiver.

Advantages of Stop and Wait Flow Control:

- Simplicity: It is a straightforward method to implement.
- Accuracy: This method provides accurate information since each frame is individually checked.
- Reliability: It ensures that data is reliably transmitted without the risk of data loss.

Disadvantages of Stop and Wait Flow Control:

- Slow Transmission: This method operates at a slow pace because it waits for acknowledgment before sending the next frame.
- Low Efficiency: Only one frame can be sent at a time, which reduces the efficiency of data transmission.
- Increased Latency: The constant waiting for acknowledgment increases latency in the transmission process, which may not be suitable for applications requiring low latency.

2. Sliding Window Flow Control: In this method, the sender can send multiple frames to the receiver without waiting for acknowledgments.

- Both the sender and receiver have fixed-size buffers called windows.

Advantages of Sliding Window Flow Control:

- Improved Performance: It performs better compared to the stop-and-wait method, as it allows for a more efficient use of network resources.
- Increased Efficiency: This method increases the efficiency of data transmission by allowing multiple frames to be in transit simultaneously.
- Higher Throughput: It can transmit many frames concurrently, leading to higher throughput.

Disadvantages of Sliding Window Flow Control:

- Handling Complexity: Managing the transfer of multiple frames in this method can be complex and requires additional mechanisms for tracking and retransmitting lost frames.
- Sequence Dependency: The receiver must receive the data frames in the correct sequence, which can be challenging in case of out-of-order arrivals or lost frames.

Random Access Protocols:

- It is a set of rules that allows stations in the network to detect and avoid traffic.
- In this, all stations have same superiority that is no station has more priority than another station.
- Any station can send data depending on medium's state (idle or busy).

It has two features:

There is no fixed time for sending data.

There is no fixed sequence of stations sending data.

Types of Random Access Protocols:

ALOHA:

- It was designed for wireless LAN but is also applicable for shared medium.
- In this, multiple stations can transmit data at the same time and can hence lead to collision and data being garbled.

Types of ALOHA as below:

1. **Pure ALOHA:**
    - When a station sends data it waits for an acknowledgement.
    - If the acknowledgement doesn't come within the allotted time then the station waits for a random amount of time called back-off time (Tb) and re-sends the data.

- Since different stations wait for different amount of time, the probability of further collision decreases.

Advantages of Pure ALOHA:

- The main advantage of pure ALOHA is its simplicity in implementation.
- It adapts to varying number of stations.
- It is superior to fixed assignment when there are a large number of bursty stations.

Disadvantages of Pure ALOHA:

- Its performance becomes worse as the data traffic on the channel increases.
- At high loads, collisions are very frequent.
- It requires queuing buffers for retransmission of packets.
- Theoretically proven throughput maximum of 18.4%.

Slotted ALOHA:

- It is similar to pure aloha, except that we divide time into slots and sending of data is allowed only at the beginning of these slots.
- If a station misses out the allowed time, it must wait for the next slot.
- This reduces the probability of collision.

Advantages of Slotted ALOHA:

- The big advantage of Slotted ALOHA is the increase in channel utilization.
- Simple to implement.
- It doubles the efficiency of ALOHA.

Disadvantages of Slotted ALOHA:

- It requires queuing buffers for retransmission of frames.
- Synchronization required.

b. CSMA:

- Carrier Sense Multiple Access ensures fewer collisions as the station is required to first sense the medium (for idle or busy) before transmitting data.
- If it is idle then it sends data, otherwise it waits till the channel becomes idle.
- However there is still chance of collision in CSMA due to propagation delay.

There are three main approaches to CSMS protocols as below:

- **1-Persistent CSMA:** The node senses the channel, if idle it sends the data, otherwise it continuously keeps on checking the medium for being idle and transmits unconditionally(with 1 probability) as soon as the channel gets idle.

- **Non-Persistent CSMA**: The node senses the channel, if idle it sends the data, otherwise it checks the medium after a random amount of time (not continuously) and transmits when found idle.
- **P-Persistent CSMA**: The node senses the medium, if idle it sends the data with p probability. If the data is not transmitted ((1-p) probability) then it waits for some time and checks the medium again, now if it is found idle then it send with p probability. This repeat continues until the frame is sent. It is used in Wi-fi and packet radio systems.
- **CSMA (Collision Detection (CSMA/CD)):** It is used to detect collisions on the network which happen when two stations attempt to transmit at the same time. CSMA/CD is a set of rules which determines how network devices respond when two devices attempt to use a data channel simultaneously.

Ethernet

- Ethernet is communication protocol which is used to connect computer in different areas.
- Ethernet is LAN technology that is used to control the flow of information from sender to receiver.
- It was developed by **Xerox** in 1970 and that time it was speed around 2,3 MB.
- Ethernet standard IEEE802.3.

Advantage of Ethernet

- Provide fast speed
- High security
- Maintain and troubleshoot easy.
- Low cost
- Switch and hub not use
- Each device has equal responsibility and privileges for sending and receiving data.

Disadvantage of Ethernet

- Used in small area
- If traffic increase on transmission media, the efficiency of Ethernet get low
- After receiving message, receivers do not provide any acknowledgement.

Types of Ethernet

1. **Fast Ethernet:** This type of Ethernet network uses cables called twisted pair or CAT5. It can transfer data at a speed of around 100 Mbps (megabits per second).
   - Fast Ethernet uses both fiber optic and twisted pair cables to enable communication.
   - There are three categories of Fast Ethernet: 100BASE-TX, 100BASE-FX, and 100BASE-T4.
2. **Gigabit Ethernet:** This is an upgrade from Fast Ethernet and is more common nowadays.
   - It can transfer data at a speed of 1000 Mbps or 1 Gbps (gigabit per second).
   - Gigabit Ethernet also uses fiber optic and twisted pair cables for communication.
   - It often uses advanced cables like CAT5e, which can transfer data at a speed of 10 Gbps.
3. **Switch Ethernet:** This type of network involves using switches or hubs to improve network performance.
   - Each workstation in this network has its own dedicated connection, which improves the speed and efficiency of data transfer.
   - Switch Ethernet supports a wide range of speeds, from 10 Mbps to 10 Gbps, depending on the version of Ethernet being used

Token Ring

- A token ring is a data link for a local area network (LAN) in which all devices are connected in a ring or star topology and pass one or more tokens from host to host.
- A token is a frame of data transmitted between network points. Only a host that holds a token can send data, and tokens are released when receipt of the data is confirmed.
- IBM developed token ring technology in the 1980s as an alternative to Ethernet
- Token Ring Is IEEE 802.5 Standard.
- Used Ring Topology
- It Used Access Control Method Token Passing
- Token Ring Is Unidirectional
- The data rate of Token Ring is 16 Mbps.

Advantages of token rings

- Token rings reduce the chances of data collision.
- Token passing performs better than bus topology under heavy traffic.
- A server is not needed to control connectivity among the nodes.

Disadvantages of token rings

- Data packets must pass through all the nodes, slowing routing.
- Token ring networks have suffered from network management problems and poor network fault tolerance.
- **Difference between FDDI and Token Ring:**

FDDI	Token Ring
FDDI is a backbone to connect LANs.	Workstations are connected directly.
The data rate of FDDI is 100 Mbps.	The data rate of Token Ring is 16 Mbps.
Stations may generate a new token.	Stations may not generate a new token.
Fiber optic cables are used.	Shielded Twisted Pair cables are generally used.
It can support large bandwidth.	It can support small bandwidth.
Workstations are interconnected by dual fiber link.	Workstations are interconnected by single ring network.
Several frames on the ring at the same time.	Only one frame at a time on the ring.

Bluetooth:

- It is a wireless technology used to transfer data between different electronic devices.
- The distance of data transmission is small in comparison to other modes of wireless communication.
- Bluetooth develops wireless connections between the laptops, handheld PCs, wireless communication devices and mobile phones.
- Bluetooth network uses short range radio technology to create short range networks.
- A fundamental Bluetooth wireless technology strength is the ability to simultaneously handle both data and voice transmissions.
- This enables users to enjoy a variety of innovative solutions such as hands-free headset for voice calls, printing and fax capabilities etc.

The Key features of Bluetooth Technology are:

- Less Complicated.
- Less Power consumption.
- Available at cheaper rates.
- Robustness.

Architecture of Bluetooth: The architecture of Bluetooth defines two types of networks:

a. Piconet:

- A Small Bluetooth network is called as Piconet.
- It uses a master/slave model to control when and where devices can send data.
- In this model, a single master device can be connected to up to seven different slave devices.
- The Master coordinates communication throughout the piconet.
- It can send data to any of its slaves and request data from them as well.
- Slaves are only allowed to transmit to and receive from their master.
- The communication between master and slave may be one to one or one to many communications.

b. Scatternet:

- It is formed by combining various piconets.
- A device can be a master of only one piconet.
- A slave of a piconet can act as a master in other piconet.
- In this way, it receives all messages from the master of its piconet and delivers packets to its slaves in other piconet.

Applications of Bluetooth:

- Wireless mouse and keyboard can communicate with the system unit.
- Modern health care devices send signals to monitors in wireless manner.
- Almost all the latest mobiles, palmtops use this technology to transfer data rapidly.
- The transfer of files, images and MP3 between mobile phones.
- Video game systems are using this technology as well as for their wireless controllers.

VLAN

A VLAN, or Virtual Local Area Network, is a network technology that allows you to create logical, segmented networks within a physical network infrastructure.

VLANs are used to improve network performance, security, and manageability by dividing a single physical network into multiple isolated virtual networks.

Each VLAN operates as if it were a separate physical network, even though it shares the same physical infrastructure.

VLANs offer several features and benefits, including:

Improved network security: VLANs can be used to separate network traffic and limit access to specific network resources. This improves security by preventing unauthorized access to sensitive data and network resources.

Better network performance: By segregating network traffic into smaller logical networks, VLANs can reduce the amount of broadcast traffic and improve network performance.

Simplified network management: VLANs allow network administrators to group devices together logically, rather than physically, which can simplify network management tasks such as configuration, troubleshooting, and maintenance.

Flexibility: VLANs can be configured dynamically, allowing network administrators to quickly and easily adjust network configurations as needed.

Cost savings: VLANs can help reduce hardware costs by allowing multiple virtual networks to share a single physical network infrastructure.

Scalability: VLANs can be used to segment a network into smaller, more manageable groups as the network grows in size and complexity.

Advantages of VLAN's:

- Reduction in cost.
- Reduction in time.
- Increased Security.
- Increased Performance.

FDDI:

- Its full form is "Fiber Distributed Data Interface".
- It provides a standard for data transmission in a local area network.
- It is a high performance fiber optic token ring.
- It is highly reliable way to transmit data using a dual-ring protocol based on the token ring protocol.
- A token ring network is a local area network where all the computers on the network are connected in a giant circle and a token-passing setup is employed to prevent data collision when two computers send data at the same time.
- FDDI covers large geographical areas.
- It can support thousands of users on the local area network.
- A FDDI network contains two fiber rings, one transmitting clockwise and the other transmitting counter clockwise.

Objectives of FDDI:

- Highly powerful workstations and servers.
- Intensive network applications.
- Larger spans of distributed network.
- Higher capacity.
- Longer distance loops.
- High performance with simultaneous transactions.

Usefulness of FDDI:

- In applications which require high security.
- As network connecting high speed computers.
- As a backbone network with high traffic levels.
- In applications which require a high degree of fault tolerance.

Wireless LAN:

- A wireless LAN uses wireless transmission medium for data transfer operation.
- A wireless local area network (WLAN) is a wireless distribution method for two or more devices that use high-frequency radio waves and often include an access point to the Internet.
- A WLAN allows users to move around the coverage area, often a home or small office, while maintaining a network connection.
- A WLAN is sometimes call a local area wireless network (LAWN).

Characteristics of WLAN

- Seamless operation.
- Low power for battery use.
- Simple management, easy to use for everyone.
- Protection of investment in wired networks.
- Robust transmission technology.

Advantages of WLAN

- Installation speed and simplicity.
- Installation flexibility.
- Reduced cost of ownership.
- Reliability.
- Mobility.
- Robustness.

Disadvantages of WLAN

- Slower bandwidth.
- Security for wireless LANs is the prime concern.
- Less capacity.
- Wireless networks cost four times more than wired network cards.

Network Hardware Components:

1. **Connectors:**
   - Connectors are used to connect network cables to terminals or other devices.
   - The connectors serve as the physical connection between the computer and the cable media.
   - Common network connections require cable media and special connectors to link the medium.
2. **Transceivers:**
   - The Transceivers is a device comprising both a transmitter and a receiver which are combined and share common circuitry or a single housing.
   - When no circuitry is common between transmit and receive functions, the device is a transmitter-receiver.
   - Transceivers are devices used in Ethernet networks to connect nodes to the physical medium.
   - They serve as both the physical connection and the electrical interface between a node and the physical medium, enabling the mode to communicate with the medium.
3. **Repeaters:**
   - A repeater is a device that receives a digital signal on an electromagnetic or optical transmission medium and regenerates the signal along the next leg of the medium.
   - In electromagnetic media, repeaters overcome the attenuation caused by free-space electromagnetic-field divergence or cable loss.
   - A series of repeaters make possible the extension of a signal over a distance.
   - Repeaters remove the unwanted noise in an incoming signal.

   Advantages of Repeaters:

   - The repeaters are very simple to implement.
   - They are very cheap in cost.
   - They allow easy expansion of the network.
   - They have little propagation delay.

Disadvantages of Repeaters:

- They cannot connect two dissimilar networks.
- They cannot filter frames.
- They do not reduce network traffic.

4. **Hubs:**

- Hub is a device that splits a network connection into multiple computers.
- It is like a distribution center.
- When a computer requests information from a network or a specific computer, it sends the request to the hub through a cable.
- The hub will receive the request and transmit it to the entire network.
- Each computer in the network should then figure out whether the broadcast data is for them or not.

5. **Network Interface Card:**

- Each computer in a network has a special expansion card called a network interface card (NIC).
- The NIC prepares (formats) and sends data, receives data, and controls data flow between the computer and the network.
- On the transmit side, the NIC passes frames of data on to the physical layer, which transmits the data to the physical link.
- On the receiver's side, the NIC processes bits received from the physical layer and processes the message based on its contents.

6. **Switches:**

- Switch is a telecommunication device grouped as one of computer network components.
- Switch is like a Hub but built in with advanced features.
- It uses physical device addresses in each incoming messages so that it can deliver the message to the right destination or port.
- Like a hub, switch doesn't broadcast the received message to entire network, rather before sending it checks to which system or port should the message be sent.
- In other words, switch connects the source and destination directly which increases the speed of the network.

7. **Routers:**

- When we talk about computer network components, the other device that used to connect a LAN with an internet connection is called Router.

- When you have two distinct networks (LANs) or want to share a single internet connection to multiple computers, we use a Router.
- In most cases, recent routers also include a switch which in other words can be used as a switch.
- You don't need to buy both switch and router, particularly if you are installing small business and home networks.

8. PC Cards:

- A PC Card is a standard for the peripheral interface for laptop / notebook computers.
- PC Cards are defined and developed by the Personal Computer Memory Card International Association (PCMCIA), a group of industry organizations in the U.S.
- PC Cards are the credit-card sized devices that enhance and extend the functions of a portable PC.
- PC Cards are no longer commonly used for memory expansion.

9. Gateway:

- A Gateway joins two dissimilar networks.
- It operates at application layer of ISO-OSI reference model.
- Gateway is exit or entrance point to a network and performs changes in protocols if required.
- Gateways are also called protocol converters and can operate at any network layer.
- Gateways are generally more complex than switches or routers.
- A gateway is also called a protocol converter.

10. Bridge:

- A bridge is a type of computer network device that provides interconnection with other bridge networks that use the same protocol.
- Bridge devices work at the data link layer of the Open System Interconnect (OSI) model, connecting two different networks together and providing communication between them.
- Bridges are similar to repeaters and hubs in that they broadcast data to every node.

- However, bridges maintain the media access control (MAC) address table as soon as they discover new segments, so subsequent transmissions are sent to only to the desired recipient.
- Bridges are also known as Layer 2 switches.